REQUIREMENT ANALYSIS DOCUMENT

Online Party Arrangement Service System

IT 4005

1. Introduction

1.1 project background

EventBridge is a full-service event coordination company that does not own physical venues; clients host events at locations of their own choosing. EventBridge's value lies in three areas: an in-house inventory of rental equipment (tents, lighting, sound, furniture, staging) that it owns and delivers itself; a curated network of external vendors (caterers, DJs/entertainers, decorators, photographers); and planning/on-site coordination, including a staff coordinator for larger events.

The business currently runs on shared spreadsheets, WhatsApp groups, and phone calls, which has led to equipment double-bookings, slow multi-vendor coordination, inconsistent custom quoting, and no real-time visibility into event-day issues.

1.2 Scope

The proposed system covers four principal areas:

- A customer-facing portal for browsing packages, customizing events, requesting custom quotes, checking real-time availability, and paying online.
- An admin/owner dashboard for managing packages, in-house inventory, the vendor network, staff/coordinators, pricing, quote routing, and reporting.
- A vendor portal for external partners to manage their own availability, respond to quote requests, and track assigned jobs.
- An event coordinator module for day-of checklist tracking and real-time issue flagging.

The scope excludes vendor payroll/accounting beyond payment tracking relevant to bookings, and does not include venue management, since EventBridge does not own physical spaces.

2. Business Overview (As-Is Analysis)

2.1 Business Description

EventBridge coordinates parties and events end-to-end without owning venues. It supplies in-house rental equipment, coordinates external vendors, and provides on-site coordination for larger events.

2.2 Current Operational Workflow

Equipment allocation is tracked on a shared spreadsheet, updated inconsistently by staff working from memory, which has caused the same equipment to be promised to two different events on overlapping dates.

Vendor coordination happens over WhatsApp: for a single event requiring a caterer, DJ, and photographer, staff must check each vendor's availability individually, often losing vendors to other jobs while waiting on replies.

Custom quotes are priced ad hoc over the phone, with no consistent process for gathering or comparing competing vendor prices for the same job.

Event-day coordination relies on printed checklists; coordinators have no real-time way to flag problems such as a late vendor or missing equipment.

2.3 Problems Identified

- Equipment double-booking from a manually updated, error-prone spreadsheet.
- Slow, sequential vendor availability checks that risk losing vendors to competing jobs.
- No structured process for gathering or comparing competing vendor quotes on custom requests.
- No real-time visibility into event-day issues for admin to respond to.
- No package structure, so every quote is built from scratch.
- No tracked history of vendor reliability, popular packages, or repeat customers.

3. Stakeholders and User Roles

The system must support the following user roles, each with a distinct set of permissions and responsibilities:

Role	Description
Customer	Browses packages, customizes events, requests and compares custom quotes, books and pays online, and tracks event status.
Admin / Owner	Manages packages, in-house inventory, the vendor network, staff/coordinators, pricing, quote routing, and reporting.
Vendor	Publishes own availability, receives and responds to quote requests, views assigned jobs, and updates job status.
Event Coordinator (Staff)	Assigned to a specific event; tracks the day-of checklist and flags real-time issues to admin.

4. Proposed System Overview

The proposed system replaces spreadsheets, phone calls, and chat threads with a centralized platform providing real-time inventory and vendor availability, structured tiered packages with add-ons, a competitive vendor quote/bidding workflow for custom requests, and live event-day coordination. Core design principles established during requirement gathering:

- In-house inventory items are tracked as a scarce, shared resource with reservation windows spanning delivery through pickup, not just event hours — preventing the double-booking problem directly.
- Custom requests are routed to multiple matching vendors per category simultaneously, replacing sequential phone/WhatsApp checks with parallel, time-boxed bidding.
- A package-plus-add-on booking model lets clients start from a standardized tier (Basic/Standard/Premium) and layer on extras, keeping pricing transparent.
- A defined inquiry-to-booking lifecycle (Inquiry → Availability Check → Provisional Hold → Confirmed Booking) replaces ad hoc coordination with a trackable pipeline, mirroring the same hold-and-lock mechanism used for inventory and vendor slots.
- Event coordinators get a live, package-derived checklist and a real-time issue-flagging channel to admin, replacing printed checklists.

5. Functional Requirements

Functional requirements are grouped by module and tagged with a unique ID and MoSCoW priority.

5.1 Customer Portal

Requirement
Login with email and a preferred password
Browse package tiers (Basic, Standard, Premium) with inclusions and indicative pricing.
Customize an event: guest count, date, location, and add-on selection.
View real-time availability of required in-house inventory and relevant vendor categories for the selected date.
View a live price calculator that updates as the package and add-ons are customized.
Select add-ons from the catalog (extra hours, upgraded lighting/sound, extra catering headcount, live food stations, security/valet, extra photography hours, specialty dietary catering).
Specify catering guest-count breakdown by dietary type (vegetarian / non-vegetarian / vegan / other).
Submit a Custom Package quote request specifying event type, guest count, date, location, and desired vendor categories.
View and compare competing vendor bids received for a custom quote request.
Select a vendor per category from submitted bids.
Place a booking that results in a provisional 48-hour hold on required inventory and vendor slots.
Complete online payment (full or partial/deposit) via payment gateway, with automatic receipt generation.
Track booking status through defined stages: Pending → Confirmed → In Progress → Completed → Cancelled.
Receive automated notifications: booking confirmation, payment receipt, reminders, and post-event follow-up.
Access booking history via an optional lightweight account (guest checkout available by default).
Submit a review/rating per vendor after event completion.
Submit cancellation or rescheduling requests governed by EventBridge's defined refund/deadline policy.

5.2 Admin / Owner Dashboard

Requirement
Manage package tiers: create, edit, price, and activate/deactivate.
Manage the add-on catalog.
Manage in-house inventory: item categories, quantities, per-date reservations, low-stock and maintenance status.
Manage vendor profiles: category, contact details, availability windows, and performance history.
Route custom quote requests to 2–3 matching vendors per requested category and track bid responses within a defined response window.
View a central calendar across inventory and vendor allocations with built-in conflict detection.
Manage staff/coordinators: assign to specific bookings and track assignment.
Record and track all payments (online and manual), outstanding balances, refunds, and vendor payouts.
Generate reports: revenue by package tier, inventory utilization, vendor performance, quote-to-booking conversion rate, and repeat-customer rate.

Requirement
Create or edit bookings manually on behalf of phone-in customers.
Maintain an event history archive of past arrangements as reusable templates.
Convert a provisional hold into a confirmed booking upon payment, automatically locking inventory and vendor assignments and generating the dietary/headcount breakdown for the caterer.
Resolve coordinator-flagged, real-time event-day issues (e.g., assigning a backup vendor).
Configure loyalty/discount rules for repeat customers.

5.3 Vendor Portal

Requirement
Vendor login to publish and update their own availability calendar.
Receive quote requests and submit a bid (price and notes) within the defined response window.
View assigned/confirmed jobs and associated event details.
Update job status (accepted, in progress, completed).
View own performance history and ratings.

5.4 Event Coordinator Module

Requirement
View assigned events and an auto-generated day-of checklist derived from the selected package and add-ons.
Check off checklist items as completed.
Flag real-time issues (late vendor, missing equipment) that immediately notify admin.
Mark an event In Progress or Completed, triggering inventory teardown/pickup scheduling.

5.5 Booking & Quote Lifecycle

Requirement
Enforce the booking lifecycle: Inquiry → Availability & Compatibility Check → Provisional Hold (48 hours) → Confirmed Booking & Advance Payment.
Enforce the quote lifecycle: Requested → Quoted → Selected → Confirmed, or Expired if no bids arrive within the response window.
Automatically release an expired provisional hold back into available inventory and vendor slots.
Automatically expire quote requests that receive no vendor bids within the response window.
Log every booking, payment, bid, and status change with a timestamp for audit purposes.

6. Non-Functional Requirements

Non-functional requirements define the quality attributes the system must exhibit.

Attribute	Requirement
Availability / Reliability	Real-time inventory and vendor-slot availability with a locking mechanism at time of booking, preventing double-allocation under concurrent access.
Security	Encrypted storage for sensitive data; payment card data is never stored directly — a PCI-compliant gateway is used; secure authenticated login for admin, vendor, coordinator, and customer roles.
Scalability	The system must support growth in booking volume, inventory catalog size, and vendor network without structural redesign.
Usability	The customer-facing portal must be simple for non-technical users and mobile-friendly, since most customers browse on phones.
Auditability	Every booking, payment, bid, and status change is logged and timestamped.
Notification Infrastructure	Reliable email and/or SMS integration to support automated confirmations, reminders, and follow-ups.
Performance	Availability checks, the live price calculator, and quote-bid comparisons must respond quickly enough to feel instantaneous.

7. Business Rules

- In-house inventory reservations cover the full delivery-to-pickup window, not just the event hours, so setup/teardown time is protected from double-booking.
- A package is a bundle definition (Basic, Standard, Premium); Custom requests are not fixed-price and route through the quote/bidding workflow instead.
- A provisional hold automatically expires 48 hours after being placed if no deposit or full payment is received; inventory and vendor slots are released back to availability.
- A quote request automatically expires if no vendor bids are received within the defined response window (e.g., 72 hours).
- A booking is only confirmed — and inventory/vendor slots only locked — once a deposit or full payment is recorded.
- A dedicated event coordinator is assigned only to Standard (partial check-in) and Premium (full-day) package bookings; Basic bookings receive delivery-crew support only.
- Reviews are tracked per vendor and feed into vendor performance history, which is used when matching vendors to future quote requests.
- Customers with two or more completed bookings become eligible for loyalty/repeat-customer discount pricing, configurable by the admin.
- Add-ons are layered on top of a base package as separate line items and must appear individually on the invoice and on the coordinator's day-of checklist.
- Cancellations and reschedules follow EventBridge's defined refund and deadline policy, configurable by the admin.

8. Key Data Entities (High-Level)

The following core entities were identified during analysis. A full entity-relationship diagram and normalized schema will be developed in the design phase; this table summarizes the entities and their purpose.

Entity	Purpose
Customer	Account holder (optional) or guest, with contact details and booking history.
Package	A tier definition (Basic/Standard/Premium/Custom) with inclusions, duration, and base price.
Add_on	An optional item/service selectable on top of a package.
Inventory_Item	A physical asset owned by EventBridge (tent, lighting rig, furniture, etc.) with quantity on hand.
Inventory_Reservation	Links an inventory item to a booking for a specific date range (delivery through pickup).
Vendor	External service provider (category, contact info, availability, performance history).
Vendor_Availability	Vendor-published open/blocked dates.
Quote_Request	A custom request describing event details and desired vendor categories.
Vendor_Bid	A vendor's price/notes submitted in response to a quote request.
Booking	A confirmed or in-progress reservation: package, add-ons, date, guest count, status, and lifecycle timestamps.
Booking_Vendor_Assignment	Links a vendor to a specific booking with job status.
Coordinator	EventBridge staff member available for event-day assignment.
Checklist_Item	A day-of task derived from a booking's package/add-ons, with completion status.
Payment	A recorded transaction (amount, method, status, linked booking, timestamp).
Review	A post-event rating/feedback record linked to a booking and a specific vendor.

9. Assumptions and Constraints

9.1 Assumptions

- EventBridge does not own venues; all bookings assume the client supplies their own event location.
- Vendors are independent partners, not employees, and manage their own calendars within the system.
- Customers have reliable access to smartphones or computers with internet access for the self-service portal.
- A third-party PCI-compliant payment gateway will be used rather than custom card-data storage.

9.2 Constraints

- This is being developed as a course/assignment project, so scope and timeline are bounded by academic deliverable deadlines (approximately 4 weeks).
- Integration depends on the availability and terms of external services (payment gateway, SMS/email providers).
- The system must remain usable by non-technical staff, vendors, and customers, limiting UI complexity.